

Remote Learning and Educational Inequality

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Abstract

Educational equity serves as a cornerstone for a fair and flourishing society. However, existing research consistently highlights the presence of educational disparities between developed and developing regions. Remote learning, a transformative educational approach altering knowledge dissemination and access, takes center stage. Our study delves into the potential of remote learning to bridge this educational divide. Analyzing students' performance in China's National College Entrance Examination from 2018 to 2020, we exploit the pandemic-induced shift to remote learning as an external catalyst. Employing a difference-in-differences methodology, we gauge remote learning's impact on performance gaps across regions. Our findings illuminate a marked enhancement in learning outcomes for students in developing regions, surpassing their counterparts in developed ones. Furthermore, we unearth that the digital divide in access to technological resources constrains remote learning's efficacy in reducing educational disparities. Our study underscores remote learning's implications and underscores the pivotal role of digital technology capabilities.

Keywords

Remote learning, Educational inequality, Digital divide, Natural experiment

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1 Introduction

Educational equity measures whether people in different groups or conditions receive equal educational opportunities and resources. A strong foundation for a fair and thriving society, educational equity promotes economic growth, social mobility, and innovation (Corak, 2013; Nickell, 2004). Despite efforts to ensure equity through digital tools, such as online crowdfunding (Keppler et al., 2022) and open online education platforms (Zhang et al., 2017), educational resources remain unequally distributed across races, genders, and regions. Particularly, educational inequality between developed and developing regions has been repeatedly identified (Sterlacchini, 2008; Xie and Zhou, 2014). Schools in developing regions often suffer from inadequate and inferior educational resources, including insufficient teaching budgets, outdated teaching facilities, inexperienced teachers, and uncompetitive peers (Yang et al., 2014). This hinders students' learning and impedes their pursuit of higher education and employment opportunities. Consequently, addressing educational inequality across regions is a pressing global issue.

Remote learning, predominantly an online educational approach encompassing synchronous, real-time peer interactions and asynchronous, self-paced learning activities

independent of instructors (Graham, 2006), is facilitated through emerging technologies like teleconferencing tools (e.g., Zoom) and online course platforms (e.g., Coursera). This transformation integrates technology into teaching and learning, revolutionizing students' access to educational resources and their interactions with peers and teachers, which together influence learning outcomes. In the traditional brick-and-mortar learning paradigm, students adhere to rigid schedules and predefined learning processes governed by schools. Educational resources provided by schools largely dictate their learning outcomes, leading to inequality between schools in developed and developing regions.

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However, remote learning introduces two distinctive aspects that hold the potential to ameliorate this disparity. Firstly, it overcomes geographical and physical constraints, granting students flexible and cost-effective access to online resources. Secondly, it empowers students with greater control over learning schedules and the flexibility to focus on personalized learning needs (Gunawardena and McIsaac, 2013).

These attributes of remote learning confer particularly advantageous outcomes to students in developing regions, affording them benefits that surpass those of their counterparts in developed regions. Specifically, the challenges of learning from skilled and accomplished yet non-local instructors, which used to be arduous and financially taxing, are now surmountable through the virtual classroom, accessible from any location with internet connectivity. This expansion significantly broadens the spectrum of instructors available to students in developing regions, tapping into a diverse pool of talent. Online forums further facilitate access to a plethora of tutorials and peer interactions for students in developing regions, granting them the autonomy to engage as per their convenience (Bhatt, 2002). By enabling students from developing regions to engage with high-achieving peers in other areas, remote learning cultivates a positive peer effect and fosters an enriched learning milieu. Furthermore, remote learning liberates students in developing regions with more discretionary time, which can be dedicated to exploring educational resources beyond their institutional confines. Given the manifold advantages, remote learning holds the potential to recalibrate the uneven dissemination of educational resources, thereby serving as a catalyst in diminishing educational inequality.

However, to fully reap the benefits of remote learning, students need to have access to digital infrastructure and possess digital skills (Ali, 2020). This study delves into the digital divide manifested across regions in terms of accessibility to and proficiency in digital technologies, encompassing hardware, software, and internet connectivity (Selwyn, 2004). In certain areas, limited internet accessibility and the absence of digital devices render online educational resources inaccessible. Furthermore, students and educators in such regions may possess limited familiarity with digital tools and resources, impeding their capacity to effectively organize, retrieve, and employ online materials. These challenges create obstacles that hinder students from optimizing their learning efficiency during remote education. While remote learning holds the promise of mitigating the educational resource disparity, realizing these advantages hinges upon the presence of digital infrastructure and competencies. In reality, in the absence of sufficient digital infrastructure and skills, remote learning could potentially amplify educational inequality.

Overall, it is important to find answers to the following questions: *Does remote learning reduce educational inequality between developed and developing regions? Does the digital divide weaken this effect?*

To address these inquiries, we capitalize on the abrupt and pervasive transition to remote learning prompted by the coronavirus 2019 (COVID-19) pandemic, which served as an exogenous shock. Before the pandemic, conventional in-person instruction predominated across most schools. However, as a response to COVID-19, stringent regulations were instituted by the Chinese government. The immediacy and urgency of these regulations led to an unprepared shift to remote learning for all students and educators. Notably, local administrations lacked the opportunity to bolster their digital technology infrastructures or provide adequate training to enhance teachers' proficiency in utilizing digital tools for online course delivery. This unique backdrop presented by the COVID-19 pandemic offers an unprecedented context to explore the potential reduction of educational inequality through remote learning, a dimension that remains unexplored. With the onset of COVID-19 in early 2020, nearly all schools in China shuttered for several months during the subsequent lockdown, compelling a transition to online teaching. High school students found themselves confined to their homes, engaging in online courses delivered by teachers through teleconferencing tools via digital devices such as smartphones and laptops. Additionally, students accessed course materials via the internet and remained connected with peers through online chat groups. As a result, remote learning became the norm for several months. Consequently, we demarcate the period before the pandemic as 2018 and 2019, and the period after as the year 2020.

Through manual curation of public declarations from school websites and press releases, we assemble a comprehensive collection of Chinese high schools' performance announcements in the National College Entrance Examination (NCEE) for the years 2018 to 2020. The NCEE, an annual standardized college entry exam administered on June 7th and 8th, is obligatory for all final-year senior high school students in China and serves as a prerequisite for admission to undergraduate institutions. Governed by the National Education Examinations Authority, the exam ensures uniformity across regions. High schools' rankings hinge significantly on students' NCEE performance, influencing their government funding. Reflecting its significance, schools strive to prepare students for this exam, which emerges as the foremost yardstick for evaluating nationwide educational outcomes. A convention in many high schools involves divulging the count of students scoring above 600 points (out of 750) in the entrance exam, a widely accepted metric of learning efficiency in China.¹ The dataset encompasses performance data from 1,458 high schools across 20 provinces from 2018 to 2020. Notably, schools in the sample adhere to a national unified test format, ensuring consistent question sets and evaluation standards, thereby alleviating potential bias arising from differing exam versions.

We use a difference-in-differences (DID) approach to estimate the impact of remote learning on educational inequality between developed and developing regions. Our analysis

reveals a substantial positive impact in 2020, with the transition to remote learning leading to a notable 24.5% surge in the number of students scoring above the 600-point threshold in developing regions as compared to their counterparts in developed regions. These findings maintain their robustness even when adopting alternative metrics such as the count of students surpassing the Yiben score or the percentage of students scoring above 600 relative to the total number of local students taking the exam. These results suggest that remote learning reduces educational inequality.²

We further investigate how the effect of remote learning is contingent on a region's digital capability. We use two variables to measure digital capability at the city level: the number of mobile phones per capita in a city and the number of broadband internet connections per capita in a city. We use the sum of two dummy variables to measure digital capability at the school level: whether a school has a website and whether that website provides a social platform account. We find that in cities or schools with a stronger digital capability, remote learning could better improve students' learning outcomes and thus more significantly reduce educational inequality.

In pursuit of uncovering the underlying mechanisms through which remote learning effectively mitigates educational inequality, we initiate an insightful survey encompassing 1,198 Chinese students who underwent remote learning in 2020. Our comprehensive questionnaire addresses diverse facets of their remote learning experience, encompassing their access to a spectrum of digital devices, their adeptness in navigating the internet and software, the reliability of their online connectivity, interactions with instructors and peers, as well as their access to a plethora of online educational resources compared to in-person learning. The findings from this survey resonate profoundly. Notably, students hailing from developing regions demonstrate an enhanced ability to connect with skilled educators and cultivate networks among high-achieving peers—an achievement facilitated by the embrace of remote learning. Furthermore, relative to their counterparts in developed regions, students in developing regions distinctly perceive elevated learning efficiency during their remote learning journey. Crucially, our survey outcomes corroborate the assertion that the efficacy of remote learning hinges on the digital capabilities of the respective region. This revelation substantiates the potential pathways elucidating our empirical findings, thus reaffirming the intricate connection between digital infrastructure, remote learning effectiveness, and the reduction of educational inequality.

We enhance our analysis by incorporating a set of rigorous robustness tests. In particular, we use the alternative outcome variable, which is the number or percentage of students surpassing the “Yiben” cutoff score, rather than the number of students exceeding 600 points of schools in Section 8.1. This is more comprehensive to represent the overall group of students' learning efficiency. Next, in Section 8.2, we adopt the matching strategy to create matched treatment and control groups that are sufficiently comparable on observable

covariates. Furthermore, in Section 8.3, we isolate resilient estimations by focusing on a balanced panel that exclusively includes schools that consistently reported their performance annually, spanning from 2018 to 2020. The outcomes consistently demonstrate strong alignment. Additionally, we execute examinations using alternate metrics (such as an alternative gage for educational resources and a metric that gages the extent of remote learning relative to the duration of lockdown) and observe a high degree of result consistency across these variations.

2 Literature Review

Our research contributes to four streams of literature: mechanism design in education, the adoption of digital technologies in education, remote learning, and the digital divide. First, our work is related to the emerging literature that addresses educational issues using operations and optimization techniques. For example, prior studies have designed various school choice mechanisms such as a lottery-plus-cutoff mechanism that matches children with schools having others from their neighborhood (Ashlagi and Shi, 2014), a public school assignment mechanism that maximizes a combination of utilitarian and max-min welfare (Ashlagi and Shi, 2016), and a two-round deferred acceptance mechanism to reassign public school seats that are vacated after the first round of centralized assignment (Feigenbaum et al., 2020). Prior research has also optimized various mechanisms to improve schools' operations, including fair and efficient course allocation mechanisms (Bichler and Merting, 2021; Budish et al., 2017), efficient bus routing algorithms (Bertsimas et al., 2019; Chu et al., 2020), how to evaluate school partners' trustworthiness using contextual factors (Keppler et al., 2021), and how to recruit students efficiently and accurately in the screening and interview stages (Du and Li, 2020).

Another key driver that can improve the quality of schools' education services is the use of digital technologies. For example, the adoption of broadband at schools enables students to access the internet faster but has been shown to have an unexpected negative impact on students' exam scores (Belo et al., 2014), and the adoption of the performance measurement system enables teachers to better track students' performance and helps improve their learning efficiency (Anderson and Kimball, 2019). Several recent influential papers by Keppler et al. (2021, 2022) study the value of online crowdfunding platforms in improving educational outcomes. For example, Gao et al. (2020) show that crowdfunding platforms can help teachers significantly improve students' academic performance using the received donations. Keppler et al. (2021) find that new technology platforms such as education crowdfunding could effectively reduce teacher turnover. Keppler et al. (2022) find that crowdfunding platforms are effective in helping teachers improve learning outcomes through the value of a small fund and that they are more effective in helping low-income schools; as a result, crowdfunding platforms can

reduce education inequality. Our work is closely related to Keppler et al. (2022) in that we both study how to reduce educational inequality using digital technologies.

We contribute to the above literature by responding to Smilowitz and Keppler (2020), who encouraged researchers to work on social issues that persist in education systems and use empirical studies to obtain insights into practice. Our work also echoes Netessine (2022) call for researchers to study socially responsible research and to contribute valuable knowledge, including accessibility to education and greater social equity. In this work, we exploit the outbreak of COVID-19 as an exogenous shock and examine the impact of remote learning on educational inequality issues. To the best of our knowledge, we are among the first to examine the social benefits of remote learning in reducing educational inequality.

In addition, our research contributes to the literature on remote learning. Prior studies have shown mixed effects of remote learning on learning outcomes. Some studies have documented the positive impacts of remote learning. For example, Goodman et al. (2019) find that remote learning can reduce the cost of accessing educational resources and therefore substantially increase overall enrollment. Cowen and Tabarrok (2014) demonstrated that remote learning can provide students with greater flexibility in where, when, and how they want to learn, which can enhance their learning outcomes. Fisher et al. (2021) demonstrated that remote learning improves sales associates' productivity because it allows them to invest their own time rather than their work time to learn about product knowledge. Researchers further point out the features of remote learning that can help improve learning outcomes and productivity by making improvements in areas such as social interactions among students (Zhang et al., 2017) and notifications regarding course assignments (Huang et al., 2021). On the other hand, some studies find negative effects of remote learning. For example, Figlio et al. (2013) randomly assigned students to in-person learning and remote learning, and they find that students achieve higher scores when they engage in in-person learning. Bettinger et al. (2017) find that remote learning does less to promote student academic success and progress in college than in-person learning. With the outbreak of COVID-19, there are emerging studies examining the effects of remote learning. Most of the evidence points to the damage of remote instruction on students' learning effectiveness (Betthäuser et al., 2023; Jack et al., 2023). However, there is also evidence that remote learning improves students' academic performance compared to students without remote learning support from school (Clark et al., 2021). Most past studies draw conclusions by examining only a single course in a single location without considering the heterogeneity of the remote learning environment in different regions, which influences educational inequality. In our study, remote learning was quickly implemented as the major learning mode during the COVID-19 pandemic over many regions worldwide. This allows us to contribute to the literature by exploring the impact of remote learning on educational inequality across

regions, which means the relative impacts of remote learning on students' academic performance in different regions.

Last but not least, we contribute to the literature on the digital divide (i.e., disparities in access to digital technologies and proficiency in using such technologies across regions and groups). The digital divide is an exemplary socio-technical phenomenon, attracting great interest from IS scholars. Previous studies show that the digital divide can be explained by regional characteristics, such as digital infrastructure (Dewan et al., 2010) and socioeconomic conditions (Hsieh et al., 2008), and by individual characteristics, such as income and education (Venkatesh and Sykes, 2013). Closing the gap in the digital divide can largely benefit social welfare. For example, prior studies have found that bridging the digital divide helps women and less connected entrepreneurs obtain venture capital funding (Wang et al., 2024). Researchers also advocate that the government should collaborate with schools to close the digital divide such as wiring schools with broadband (Belo et al., 2016). However, closing the digital divide can be challenging. For example, Anderson (2008) demonstrated that the effectiveness depends on local regions' digital capability. Bacher-Hicks et al. (2021) showed that students in regions with better digital infrastructure can better access online resources. We contribute to this stream of literature by investigating whether the digital divide might limit the effectiveness of remote learning in reducing educational inequality between developed and developing regions.

3 Theory Development

In this section, we explore potential mechanisms of how remote learning could reduce educational inequality between developed and developing regions. We then discuss how the digital divide might limit the effectiveness of remote learning in reducing educational inequality.

3.1 Educational Inequality

Traditional brick-and-mortar learning is characterized by in-person education within a fixed physical setting, where temporal and spatial factors play pivotal roles in shaping learning outcomes. In this context, face-to-face interactions between students and teachers facilitate knowledge exchange and enable progress monitoring (Staker and Horn, 2012). However, this approach demands adherence to rigid schedules and prescribed learning processes, with educational outcomes often hinging on the resources supplied by schools and educators. Geographical disparities further amplify educational inequality, with schools in developed regions typically possessing superior educational resources compared to their counterparts in developing areas.

In contrast, remote learning represents a departure from the conventional model, disentangling students, educators, and learning materials from a shared physical space. This innovative approach redefines access to resources and interactions with instructors and peers (Holmes and Gardner, 2006).

Central to remote learning are its two pivotal features: connectivity and flexibility. Firstly, it transcends geographical constraints, providing cost-effective access to instructors, peers, and digital materials (Gunawardena and McIsaac, 2013). Secondly, it empowers students with personalized learning schedules and content emphasis, adapting to individual needs (Gunawardena and McIsaac, 2013). We now delve into how remote learning can help bridge educational inequalities.

The outreach of remote learning uniquely benefits students in developing regions, promising unprecedented connections to instructors and learning resources. For these regions, remote learning facilitates engagement with accomplished educators and access to high-quality resources, which were often out of reach in traditional settings. Numerous offline courses from developed regions have transitioned to online platforms, fueling a growing reservoir of educational content (Holmes and Gardner, 2006). Additionally, the digital landscape nurtures exploration, empowering students to delve into a plethora of online resources that augment learning experiences. As a result, students in developing regions enjoy an expanded repository of enriching learning materials, bridging resource disparities.

The transformative potential of remote learning extends further, transcending geographical boundaries and financial constraints (Gunawardena and McIsaac, 2013). Accessing proficient instructors required considerable travel for students in developing regions, incurring substantial costs and time investments. Remote learning obliterates these barriers by enabling digital connections via video conferencing platforms and fostering virtual classrooms accessible from anywhere with an internet connection. This democratization of instruction amplifies the reach of talented educators, narrowing educational divides (Cowen and Tabarrok, 2014). Furthermore, digital materials, including textbooks, prove more affordable than their traditional counterparts, bolstering equitable access.

Importantly, the dynamics of peer interactions profoundly impact learning outcomes (Akşin et al., 2021; Zhang et al., 2017). In traditional settings, students' interactions are confined to their immediate classroom cohorts, favoring those in developed regions with larger pools of competitive peers (Tan and Netessine, 2019; Zeng et al., 2023). Remote learning disrupts this dynamic, affording students in developing regions unprecedented opportunities to connect with exceptional peers via online forums and platforms. This dynamic fosters a collaborative environment where diverse perspectives enrich discussions (Hwang et al., 2015). For instance, online seminars facilitate interaction and collaboration, nurturing a positive peer effect that spurs creativity and bolsters learning efficiency (Kc et al., 2020). This facet of remote learning effectively augments the learning environment for students in developing regions.

Moreover, remote learning bestows students with enhanced control and adaptability over their learning schedules (Gunawardena and McIsaac, 2013). Freed from rigid timetables and commutes, students can dedicate more time to

learning and select the most effective learning modes. Asynchronous learning opportunities, such as recorded lectures and online tutorials, complement synchronous sessions, enhancing learning versatility (Tan and Netessine, 2014). Unlike traditional classrooms, remote learners can tailor their pace and content focus, optimizing learning outcomes (Hopp et al., 2007; Song et al., 2015). This newfound flexibility not only empowers students in developing regions to access premium resources beyond local schools but also facilitates more effective learning strategies, culminating in improved outcomes.

Taken together, remote learning harnesses connectivity and flexibility to elevate learning efficacy and outcomes, with a pronounced impact on students in developing regions. While benefiting learners across the board, remote learning's transformative potential is particularly pronounced for those in developing regions, assuaging educational disparities. Thus, we posit that remote learning can serve as a powerful tool to narrow learning outcome gaps between students in developed and developing regions.

Hypothesis 1. Remote learning reduces educational inequality between developed and developing regions.

3.2 Moderating Factor: Digital Divide

In the realm of remote learning, students heavily depend on digital technologies to engage in online coursework, submit assignments, collaborate with peers and instructors, and access virtual learning resources. Thus, the presence of robust digital technology infrastructure and adept digital skills becomes pivotal for creating a conducive learning environment and ensuring a seamless remote learning experience (Bacher-Hicks et al., 2021).

The digital divide, characterized by discrepancies in digital capabilities among regions, encompasses both the availability and mastery of digital technologies (Selwyn, 2004). Regions with weaker digital prowess often grapple with issues like internet connectivity disruptions, software instability, and hardware reliability concerns (Ramdas and Sungu, 2022). The extent of the digital divide can be striking. Less developed regions frequently encounter inadequate technology infrastructure and a population grappling with limited digital literacy. For instance, in 2021, the average 4G download speed in eastern Chinese areas, including provinces like Shanghai, Jiangsu, and Zhejiang, hovered around 34.57 Mbps. In comparison, the less developed western parts of China recorded speeds of approximately 28.76 Mbps. Drawing data from the 2020 provincial Statistical Yearbooks, calculations reveal a disparity in mobile phones per household, with larger cities averaging 5.05 compared to 3.38 in smaller cities. These disparities in digital infrastructure introduce hurdles for students in such regions, impeding their learning journey. Challenges stemming from inferior digital technologies can lead to stress and frustration (Mark et al., 2008), causing distractions (Levin et al., 2012) and diminishing cognitive capacity (Kc, 2014).

Consequently, students in digitally disadvantaged areas might be deterred from accessing online content and interacting with peers and instructors from more technologically advanced regions.

Furthermore, systematic disparities extend to digital proficiency among individuals across regions. In less developed areas, people tend to possess fewer digital skills necessary for effective remote learning. In 2015, the China Internet Network Information Center reported that nearly 20% of individuals in urban areas utilized online forums for knowledge search and global interactions, contrasting with a mere 11% in less developed rural areas (China Internet Network Information Center, 2015). Deficient digital skills present barriers for students in such regions, hindering learning efficiency during remote education (Ferri et al., 2020). Remote learning mandates proficiency in using computers, operating systems, and web browsers. Additionally, competencies in evaluating and leveraging online resources are essential for content-related tasks (Pagani et al., 2016). Moreover, urban schools in China boast over 85% availability of computers and internet access for online course delivery, while rural schools lag at under 75% (Mo, 2015; Chinese Ministry of Education, 2016). Inadequate digital skills among educators can impair the delivery of engaging online courses and the creation of an effective virtual learning environment (Ferri et al., 2020). Consequently, the digital divide can adversely influence the academic outcomes of students in less developed regions.

Taking the above factors together, students in regions with deficient digital capabilities face hindrances in terms of learning efficiency in the realm of remote education. Hence, we posit that during remote learning, regions grappling with insufficient digital infrastructure and skills might witness a disproportionate impact on learning efficiency. In the event that the digital divide between developed and developing regions is substantial, the consequences of remote learning could potentially yield negative effects.

Hypothesis 2. The digital divide weakens the effect of remote learning on reducing educational inequality.

4 Data and Summary Statistics

We use Chinese high schools' NCEE results from 2018 to 2020 as our research setting and the COVID-19 pandemic-induced shift to remote learning as an exogenous shock. The NCEE is the most important test that high school students take in China. It is the standardized college entrance exam held annually in China, which is taken by all students in their last year of senior high school on June 7th and 8th every year. It is required for entrance into all higher education institutions at the undergraduate level. The entrance exam is similar to the Scholastic Assessment Test (SAT) exam in the United States. The difference is that the entrance exam is centrally designed by the National Education Examinations Authority and thus is the same for students across regions. Students' performance on this exam is the largest factor in evaluating high

schools' ranking, which affects their future funding provided by the government. Therefore, high schools in China make significant efforts to prepare students for the exam, and this exam would serve as the best context for evaluating students' learning outcomes on a large scale.

We use the average students' performance at the high school level on the entrance examination in China as the outcome variable. First, according to the information released by a city's education bureau, we collect the list of high schools in each city. Every year after the entrance examination, each school publicly announces the performance of its students. We collect the performance data from such public announcements disclosed on school websites and press releases. We gathered such announcement data from 2018 to 2020, which covers 1,458 high schools in over 200 cities from 20 provinces in China.³

These announcements report the maximum score achieved by their students and the number of students scoring above 600 points (the total score is 750 points). In the context of Chinese high school education, standardized test scores, particularly those achieved in the NCEE, hold significant weight as indicators of students' academic performance. The NCEE serves as a pivotal examination determining students' eligibility for university admissions. Against this backdrop, the number of Chinese high school students scoring above 600 points can reasonably be considered a relevant benchmark for evaluating school-level academic performance. It's important to emphasize that the 600-point threshold stands as a widely accepted criterion that reflects a student's attainment of a solid baseline academic proficiency within the context of the NCEE. This score encapsulates a student's grasp of subject knowledge, problem-solving abilities, analytical skills, and more—competencies that are valued in the Chinese high school education system. In addition, we also incorporate an alternative measure, specifically the number or percentage of students surpassing the "Yiben" cutoff score, as opposed to the number of students exceeding 600 points of schools. This alternative outcome variable is more comprehensive to represent the overall group of students' learning efficiency and we find that the results shown in Section 8.1 are highly consistent. It's important to highlight that this alternative measurement has limitations in terms of data availability.

Considering the number of students exceeding the 600-point threshold allows us to indirectly gauge a school's traditional achievements in imparting subject knowledge and cultivating academic competencies. Schools significantly contribute by providing an equitable platform for students to attain a particular academic standard, reflecting their dedication to effective instruction, capable educators, and a conducive learning environment. In effect, this metric not only highlights a school's triumph in delivering subject matter expertise but also hints at the potential success in fostering comprehensive student attributes, critical thinking, and adept problem-solving capabilities. Crucially, this approach offers a plausible means of evaluating student performance across a vast landscape

of thousands of schools spanning 20 provinces. Due to the private nature of students' entrance examination scores, maintained by each province's education bureau, directly obtaining this data from institutions would be impractical and resource-intensive.

Note that the 1,458 high schools from which we collect data account for over 30% of the total number of high schools in those cities. We acknowledge that the sample that we study is not the total sample including all schools and the sample includes schools that chose to disclose their performance. Because the percentage of schools in our sample is large enough, our studied sample can be representative of average schools. Since disclosing students' entrance exam performance is at the schools' discretion, schools might choose to publish the announcements when they perform well. As a result, our data sample before and after remote learning could contain a different set of schools, and there is a potential selection bias concern. To address this concern, we perform a balanced panel analysis in Section 8.3, where we construct a balanced panel by keeping the schools that always disclose their performance. In this way, we can compare the learning outcomes of the same group of schools before and after remote learning and find highly consistent results.

In addition, since we use the absolute number of students scoring above 600 points, a potential concern is that a change in performance might be driven by potential population migrations. If the number of students taking the entrance exam rises, an increase in the number of students scoring above 600 points does not necessarily indicate an improvement in performance. We address this concern by collecting the number of students taking the entrance examination in each city during our time window. First, we conduct the time trend analysis grouped by regions and find that the number of students taking the entrance examination in developed and developing regions follows the same time trends. It indicates that the pools of students taking the entrance examination in developed and developing regions do not have significant temporal changes, and population migrations do not drive our results. Second, we construct another dependent variable, the percentage of students scoring above 600. In Section 6.1, we use this variable to replicate the analysis of the impact of remote learning on educational inequality. The results are highly consistent.

We assign the capital city of each province as developed regions (i.e., control group) and non-capital cities in each province as developing regions (i.e., treatment group). It is a common and standard way in the literature to classify developed versus developing regions in China (Ye et al., 2021). China's central government allocates resources based on location and political considerations. Provincial capital cities are generally governed by more powerful political leaders who have the bargaining power to secure more investments, negotiate for more favorable policies from the central government, and maintain a higher level of autonomy (Burawoy, 2001; Zhou, 2000). As a result, the capital city in each province receives more investments, has better infrastructure, and has

a stronger pool of skilled labors. Therefore, provincial capital cities are typically regarded as developed regions. Furthermore, to ensure the robustness of our results, we change the definition of developing regions in Section 8.5, and we find that the results are highly consistent.

Figure 1 plots the time trends of the average number of students scoring above 600 points from schools in developed regions and developing regions from 2018 to 2020. We can observe that schools in developed regions, on average, have more students scoring above 600 points than schools in developing regions. We can also see that the performance of schools in developed and developing regions evolved in parallel before remote learning. It indicates that geographical educational inequality in China is large and persistent. After adopting remote learning in early 2020, we can see a significant increase in the learning outcomes of students in developing regions, whereas the learning outcomes of students in developed regions do not change much, which helps close the gap in educational inequality. In our paper, we adopt a DID method to rigorously identify such an impact. A visual inspection of Figure 1 shows that there are no pre-treatment differences in the score trend of developed and developing regions, which supports the parallel trend assumption.

Table 1 reports split-sample statistics before and after remote learning. No. of Students Above-600 represents the average number of students scoring above 600 points in a school. No. of Mobile Phones is the average number of mobile phones per capita in a city. No. of Broadband Internet is the average number of broadband internet connections per capita in a city. We collect the phone and internet variables from the provincial Statistical Yearbooks. We also report the statistics of a list of control variables, such as No. of Top High Schools, which is determined by the high school ranking list published by YuanFudao, the leading K-12 training company in China. The ratio of Teacher-to-Student represents the teacher and student ratio in each city, which we collect from the provincial Statistical Yearbooks. We use the migration scale index from Baidu Migration to calculate Net Population Migration, which is the number of low-skilled workers in each city. No. of Students is the average number of high-school students in a city taking the entrance exam each year, which we collect from local government reports.

We can observe that before remote learning, schools in the developed regions had an average of 166.7 students scoring above 600 points, while schools in developing regions had 121.6 such students. After adopting remote learning, the number of students above-600 in developed regions does not change much, which amounts to 156.7. However, we can see significant growth in the number of students above-600 in developing regions, which increases to 141.7. More people in developed regions have mobile phones and use broadband internet than in developing regions. In addition, we can observe that cities in developed regions have more top high schools, more students taking the entrance exam, higher

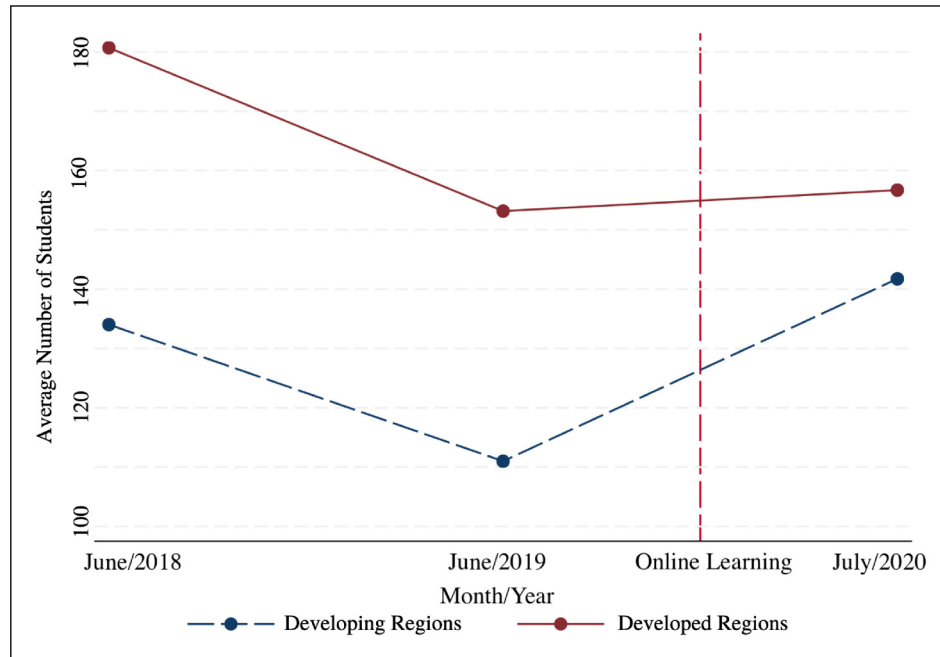


Figure 1. Time trends of students scoring above 600 points from 2018 to 2020. This graph plots the time trends of the average number of students scoring above 600 points from schools in developed and developing regions. The vertical line represents the start of the adoption of nationwide remote learning. The entrance exam in 2020 took place in July.

Table 1. Summary statistics.

Variable		Developed regions		Developing regions	
		Mean	Stdev	Mean	Stdev
Before Remote Learning	No. of Students Above-600	166.748	211.168	121.583	210.418
	No. of Mobile Phones	5.182	1.817	3.236	3.608
	No. of Broadband Internet	1.499	0.682	0.751	0.651
	No. of Top High Schools	0.162	0.369	0.068	0.251
	Ratio of Teacher-to-Student	0.088	0.015	0.079	0.015
	Net Population Migration	107.003	55.342	37.892	25.285
	Population	783.285	265.071	455.091	209.039
	No. of Students	54,371.05	16,628.76	33,137.34	17,495.78
After Remote Learning	No. of Students Above-600	156.702	212.742	141.743	211.993
	No. of Mobile Phones	5.396	1.840	3.407	3.162
	No. of Broadband Internet	1.536	0.490	0.88	0.567
	No. of Top High Schools	0.176	0.381	0.064	0.245
	Ratio of Teacher-to-Student	0.085	0.015	0.076	0.016
	Net Population Migration	106.484	54.385	37.681	25.337
	Population	780.827	271.067	464.716	220.653
	No. of Students	59,519.63	20,071.7	36,153.39	20,112.22

teacher-to-student ratios, and more net population migration than those in developing regions.

5 Identification

Our identification exploits the implementation of remote learning in response to COVID-19 as an exogenous shock that has brought about significant changes to the teaching

and learning process. This has resulted in students having to adopt remote learning at home. The validity of our approach is dependent on the assumption that the shock is exogenous to the anticipated response of schools. However, it is possible that some schools may have prepared for the shock by building their digital infrastructure and developing online teaching platforms in advance. This could confound the treatment effect. Nevertheless, the possibility of this happening is unlikely,

given the rapid development of the situation. In early January, COVID-19 was regarded as low-risk and not considered a threat to China. It was not until 23 January 2020, when Wuhan was locked down, that significant actions were taken. Within days, the quarantine was extended to all provinces and cities. On January 29th, an emergency educational policy was implemented to ensure that students could continue their learning activities by switching to large-scale online learning methods when schools were closed. Consequently, high school students in China who were preparing for their entrance exams were required to stay home and adopt remote learning during the lockdown period. Rather than providing pre-recorded courses, teachers in high school continued to teach in real-time, shifting the classroom experience to online platforms such as Dingding and VooV Meeting.

5.1 Average Treatment Effect

The DID approach is commonly used to evaluate people's or organizations' responses to natural shocks or policy change (e.g., Wan 2022; Bekkerman et al. 2023; Li and Wan 2023). Our study employs a DID analysis, utilizing the number of students who scored above 600 and the percentage of students who scored above 600 at each school as the outcome variables. The pre-treatment period encompasses the years 2018 and 2019, while the post-treatment period corresponds to the year 2020. This analysis is conducted at the school level. We compare the learning outcome gap between students in developed and developing regions before and after remote learning using the following model specification with the school-level data:

$$Outcome_{i,g,t} = c + Remote_t + Developing_g + \beta Developing_g \times Remote_t + Control + \eta_t + \delta_i + \epsilon_{i,g,t}, \quad (1)$$

where i denotes school, g denotes the region group it belongs to, and t denotes the year. We use two outcome variables: $Outcome_{i,g,t}$ represents the logged number of students scoring above 600 points for school i in region g at year t or the percentage of students scoring above 600 points for school i in region g at year t . η_t is the year fixed effect, δ_i is the school fixed effect, and $\epsilon_{i,g,t}$ is the error term. The year fixed effect η_t includes a set of year time dummies that control for the time trend. The school fixed effect δ_i includes a set of school dummies that control for differences among schools.

The dummy variable $Developing_g$ equals 1 if region g is not the capital city in a province and 0 otherwise. Its main effect is absorbed by the school fixed effects. The dummy variable $Remote_t$ equals 1 if year t is 2020 when schools adopted remote learning, and 0 otherwise. Its main effect is absorbed by the year fixed effects. The coefficient of β estimates the effect of remote learning on educational inequality, measured by the learning outcomes of students in developing regions relative to those in developed regions.

5.2 Ruling Out Alternative Explanations

The academic performance of students is influenced by multiple factors, such as the school learning environment and parental involvement. Due to the COVID-19 pandemic, students were compelled to switch to remote learning, and their parents were also required to work from home. This change may have impacted the level of parental support during the learning process. To mitigate the potential impact of other factors on educational inequality, we include a set of control variables. For instance, we account for students' baseline ability by incorporating a dummy variable denoting whether a school is a top high school, *Top High School*, which we identify using the high school ranking list published by Yuan-Fudao between 2015 and 2017.⁴ In addition, we include the teacher-to-student ratio in each city to control for educational quality. When the outcome variable is the absolute number of students scoring above 600 points, we also control the number of students taking the college entrance examination in each city.

In China, a significant proportion of the population, particularly low-skilled workers, migrates from their hometowns to larger cities for employment opportunities. Due to the restrictions imposed by the household registration system (*Hukou*), which designates a person as a permanent resident of a city and limits children to attend only the public schools in that city (Liu, 2005), these workers' children have to study in their hometowns. As a result, these workers are typically unable to stay at home and support their children's education. However, the COVID-19 pandemic altered this scenario. The outbreak coincided with the Chinese Lunar New Year, during which people returned to their hometowns to celebrate with their families. Due to the pandemic, workers were unable to return to their workplaces and had to remain at home with their children for an extended period, leading to increased time spent with their children. To account for this, we control the Migration Scale Index, which calculates the net number of low-skilled workers who enter each city divided by the number of households in that city.

Furthermore, there is another possibility that the reduced in-person learning may contribute to a narrowed gap between the NCEE performance gap between schools in developed and developing regions because in-person learning is usually more effective in developed regions than in developing regions. It is important to note that the NCEE in 2020 was postponed by one month to compensate for the loss of in-person learning caused by COVID-19 and thus was held in July instead of June, as was the case in 2018 and 2019.⁵ Since high schools returned to in-person learning before the NCEE in 2020, this postponement ensures that the increase of remote learning days in 2020 is not equal to the decrease of in-person teaching days in our setting. Hence, we construct a control variable to capture the duration of in-person teaching for students taking the NCEE each year within our time window. Controlling this variable helps us rule out such a possibility.

Lastly, some may be concerned that the Ministry of China intentionally lowered the difficulty of NCEE to alleviate the students' burden, which may dilute the advantage of schools in the developed regions over those in the developing regions. We find that at the end of 2018, the Ministry of Education, along with eight other departments, jointly issued the document "Measures for Reducing the Burden on Primary and Secondary School Students."⁶ This document represents the first comprehensive and systematic effort to reduce the academic burden on primary and secondary school students. From the end of 2018 to 2020, the time window of our study, more than 20 provinces successively introduced the implementation plan. To rule out this concern, we manually collect the specific date for each province and city to introduce the implementation plan for reducing students' burdens. According to the introduction date, we create the dummy variable to represent whether the city where the school is located introduced the plan for reducing burdens before this year. Using this dummy variable as the control variable helps address this concern.

In summary, the control variables used in the study were carefully selected to mitigate potential confounding factors affecting the relationship between remote learning and students' academic performance. These include a dummy variable for whether a school is ranked as a top high school to account for baseline student ability and the teacher-to-student ratio to control for variations in educational quality. Additionally, the Migration Scale Index was used to control for the impact of parental presence due to COVID-19 lockdowns, particularly among low-skilled migrant workers. The variable measuring the duration of in-person teaching was incorporated to address changes in the learning environment due to the pandemic's impact on the school calendar. Finally, a dummy variable indicating whether local burden reduction policies were implemented before each year was included to control for systematic differences in educational policies across regions.

5.3 Moderating Effects

The impact of remote learning on academic outcomes may not be uniform. The effect of remote learning on educational inequality is dependent on the digital capabilities of different regions. We investigate which regions are more likely to gain from remote learning. Our analysis focuses on three key moderating factors that gage the digital capacity of a city or school. To evaluate digital availability at the city level, we employ two variables: the number of mobile phones per capita and the number of broadband internet connections per capita. Mobile phones and internet connectivity represent critical components of digital infrastructure that students depend on to participate in online coursework.

Furthermore, we include the combined effect of two dummy variables to gage the digital sophistication at the school level: whether a school has a website and whether that website includes a social platform account. Websites play a crucial role in digital infrastructure by facilitating teaching,

learning, and communication (Li and Kannan, 2014; Taddeo and Barnes, 2016). Additionally, providing social platform accounts for visitors to follow the school's latest news and share opinions constitutes a vital aspect of a school's digital infrastructure, offering another channel for visitors to stay digitally connected with the school. We augment equation (1) by including the moderating factor *Digital Capability*_{*i,g*} and interacting it with the treatment effect,

$$\begin{aligned} Outcome_{i,g,t} = & c + Remote_t + Developing_g + \beta Developing_g \\ & \times Remote_t + \gamma Developing_g \times Remote_t \\ & \times Digital\ Capability_{i,g} + Control \\ & + \eta_t + \delta_i + \epsilon_{i,g,t}, \end{aligned} \quad (2)$$

where *i* denotes school, *g* denotes the regional group it belongs to, and *t* denotes the year. *Digital Capability*_{*i,g*} is the moderator, which takes one of the three values: the average number of mobile phones across all schools in region *g*, the average number of broadband internet connections across all schools in region *g*, or the digital complexity of school *i*'s website. The coefficient of interest is γ of the triple-interaction term. As before, we add school and year fixed effects and a list of control variables.

6 Empirical Results

In this section, we present the findings regarding the impact of remote learning on educational inequality. Specifically, we begin by examining the effects on the learning outcomes of students in developed and developing regions. Next, we investigate the moderating effects of the digital divide.

6.1 Effects of Remote Learning on Educational Inequality

Table 2 reports the estimated effect of remote learning on learning outcomes at the school level using equation (1). Panel A reports the results using the number of students scoring above 600 points as the outcome variable, and panel B reports the results using the percentage of students scoring above 600 points as the outcome variable. In line with our observations in summary statistics, the results show that schools in developed regions, on average, have more students scoring above 600 points than schools in developing regions. Second, after adopting remote learning, there is a significant improvement in the learning outcomes of students in developing regions relative to those in developed regions, closing the gap in learning outcomes. The coefficients of *Remote* × *Developing* across columns of Table 2 are highly consistent. We chose the coefficient of the interaction term in column 3 as our main result because it captures the finest fixed effects: school- and year-level fixed effects. The results suggest that remote learning leads to a 24.5% increase in the learning outcomes of students in developing regions relative to those in developed regions.⁷

Table 2. Effects of remote learning on educational inequality.

Panel A	Log No. of Students Above-600		
	(1)	(2)	(3)
Variables			
<i>Remote × Developing</i>	0.204*** (0.048)	0.261*** (0.048)	0.219*** (0.047)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	2,829	2,829	2,829
R ²	0.253	0.244	0.287

Panel B	Percentage of Students Above-600		
	(1)	(2)	(3)
Variables			
<i>Remote × Developing</i>	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	2,829	2,829	2,829
R ²	0.110	0.126	0.161

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

It indicates that remote learning has positive effects in reducing educational inequality.

6.2 Moderating Effects of the Digital Divide

We then proceed to assess the moderating effects of the digital divide on the influence of remote learning on educational inequality. As delineated in our theoretical framework, regions characterized by restricted digital capabilities are more susceptible to disruptions and diversions during remote learning. Furthermore, educators and students in these areas might lack familiarity with digital tools and the essential digital proficiencies needed for effective remote learning. These aspects collectively imply that the digital divide could potentially hinder the effectiveness of remote learning in mitigating educational inequality.

Table 3 reports the estimated moderating effects of the digital divide using equation (2). The coefficient of *Remote × Developing × Mobile Phones* in column 1 of Table 3 shows that remote learning can better improve the learning outcomes of students in regions with higher usage of mobile phones. The coefficient of *Remote × Developing × Broadband Internet* in column 2 shows that remote learning can better the learning outcomes of students in regions with higher broadband internet usage. Finally, the coefficient of *Remote × Developing × School Websites* in column 3 shows that remote learning can better improve the learning outcomes of students in regions where schools have higher digital complexity.

Our findings indicate that the impact of remote learning on enhancing learning outcomes is more pronounced in cities or

Table 3. Moderating effects of the digital divide.

Variables	Log No. of Students Above-600		
	(1)	(2)	(3)
<i>Remote × Developing</i>	0.118** (0.052)	0.004 (0.059)	0.188*** (0.050)
<i>Remote × Developing × Mobile Phones</i>	0.065** (0.032)		
<i>Remote × Developing × Broadband Internet</i>		0.390*** (0.137)	
<i>Remote × Developing × School Websites</i>			0.041* (0.022)
Year Fixed Effects	Yes	Yes	Yes
School Fixed Effects	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes
Observations	2,797	2,556	2,829
R ²	0.302	0.311	0.289

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). The number of observations varies in columns because the digital capability variables are missing in some cities. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

schools with higher levels of digital capability. This observation further implies that bridging the digital divide could augment the potential of remote learning to effectively diminish educational inequality between developed and developing regions.

7 Exploration of Possible Mechanisms: Evidence From a Survey

We propose three mechanisms through which remote learning could influence educational inequality. First, remote learning helps students in developing regions connect with teachers and learning resources on a broader scale and at a lower cost than in-person learning. Second, compared to in-person learning, remote learning allows students in developing regions to increase their opportunities by networking with competitive peers. Third, remote learning offers students more flexibility in their learning formats and schedules, affording students additional free time compared to the structured timetable of in-person learning.

To provide support for these mechanisms, we conducted an online survey in October 2021 (presented in Figure A.1 in the Appendix of E-Companions) with 1,776 respondents who were Chinese students in their last two years of high school in 2020. To ensure our sample is representative, we employed a stratified sampling method. We categorized universities by type and then selected schools within each category in proportion to their representation to collect data.

Table 4. Survey results.

Mechanism	Item	Developed Mean	Developing Mean	Mean Diff.
Teachers and learning resources	You use free online educational resources	3.342	3.594	−0.253***
	You use educational resources from other schools	3.303	3.561	−0.258***
	You can effectively interact and communicate with instructors	3.034	3.286	−0.252***
	Online learning resources are helpful.	3.504	3.746	−0.242***
Peers	I lack the motivation to study due to less peer pressure	3.311	3.121	0.190**
	You can effectively interact and communicate with peers	3.479	3.618	−0.139*
Free time and flexibility	Commute time between home and school	45	47	−2
Learning outcomes	You have a good learning efficiency	3.124	3.315	−0.191**

Note: This table reports the mean and mean difference between the scores rated by students in developed regions and developing regions in the survey. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To ensure randomness, we first contacted faculties who are class advisors from different levels of universities to distribute questionnaires to university students who took the college entrance examination in 2020 or 2021. The questionnaires distributed the link of this online survey to students through WeChat groups (China's most popular instant messaging tool), and students voluntarily participated. Due to the nationwide nature of China's university enrollment system, students are recruited from cities across the country, thus making the student population at different levels of universities representative to a certain extent.

In this survey, we ask students to identify the city where their high school is located to help determine whether they were in developed or developing regions during remote learning. We also ask students to measure how much they agree with various statements concerning their remote learning experience. The level of agreement is measured by a five-point semantic differential scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). We pretest the survey and confirm that the questions are unambiguous. To verify whether respondents are paying attention when filling out the questionnaire, we include one item in which we ask respondents to choose a specific option. After deleting absent-minded respondents from our sample, we are left with 1,198 valid questionnaires: 234 done by students in developed regions and 964 done by students in developing regions. The effective response rate is 67.5%. Meanwhile, by analyzing respondents' demographic information, we find that they come from over 180 cities, covering the 20 provinces in our main analysis. This breadth of coverage offers credibility for our analysis based on the survey. Additionally, we find that the NCEE score range of respondents ranges from below 300 to over 700 (full score is 750), which further indicates that the coverage of students in this survey is extensive.

The survey uses four items to evaluate whether students better connect with teachers and learning resources in a remote learning environment compared to in-person learning: (1) whether the student has more opportunities to use free online

educational resources; (2) whether the student has more opportunities to use educational resources from other schools; (3) whether the student can effectively interact and communicate with instructors; (4) whether online learning resources are helpful to the student's learning. For each item, we compute the mean for students in developed and developing regions, respectively, and we test the mean difference using the analysis of variance. Table 4 shows that students in developing regions have significantly higher scores on these four items than students in developed regions. It provides support for our first mechanism. Furthermore, in Table A.9 in the Appendix of E-Companions, we include additional information about our survey results.

The survey uses two items to evaluate how students network with their peers: (1) whether the student lacks the motivation to study during remote learning because he/she cannot observe peers' learning progress; (2) compared to in-person learning, whether the student can effectively interact and communicate with peers. Table 4 illustrates that students in developing regions exhibit notably higher scores on these two aspects compared to their counterparts in developed regions. This observation lends support to our second hypothesis, indicating that students in developing regions perceive a greater ability to engage, communicate, and learn effectively from their peers during the period of remote learning compared to students in developed regions.

Furthermore, we included a single item in our survey to assess the amount of time students saved by eliminating their daily commute between home and school. The results, as presented in Table 4, reveal that students in developed regions spent an average of 45 minutes per day on a round-trip commute before the implementation of remote learning, while that number for students in developing regions is 47 minutes. This finding underscores that remote learning provides students with additional free time, enabling them to make better use of the flexibility inherent in learning formats and schedules. Lastly, we inquired about students' perceptions of their learning efficiency during remote learning. As indicated in

Table 5. Effects of the digital divide according to the survey.

Variables	Students' Learning Efficiency			
	(1)	(2)	(3)	(4)
Availability and Proficiency	0.243*** (0.037)			
Internet Connectivity		0.255*** (0.031)		
Smooth Software Usage			0.264*** (0.031)	
Capability of Teachers				0.363*** (0.034)
Control Variables	Yes	Yes	Yes	Yes
Observations	1198	1198	1198	1198
R ²	0.049	0.073	0.076	0.101

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 4, students in developing regions recorded significantly higher scores compared to their counterparts in developed regions. This observation further bolsters Hypothesis 1 and aligns with our core findings, reinforcing the notion that students in developing regions reap more substantial benefits from remote learning. Consequently, remote learning holds the potential to mitigate educational inequality.

Recall that we also show that the effectiveness of remote learning is contingent on regional digital capabilities. In this survey, we use four items to measure digital capabilities that students experience during remote learning: (1) whether the student can access digital devices such as mobile phones and computers and proficiently use them to take online courses; (2) whether the student experiences any internet connectivity issues when taking online courses; (3) whether the student has a smooth software usage experience; (4) whether teachers use digital devices proficiently to deliver online courses. Then, we use students' scores on NCEE and the dummy variable representing whether the student comes from the developing regions as control variables. By regressing the learning efficiency score on these digital capacity scores separately, we obtain the results in columns 1 through 4 in Table 5. Note that adding the two control variables helps address the possibility that students with higher NCEE scores or from developing regions might be more inclined to perceive the benefits of remote learning. The results in Table 5 indicate that students with better and stronger digital capability show better learning efficiency, which provides additional support for Hypothesis 2. Moreover, we also use students' scores on NCEE as an alternative dependent variable that could better capture students' actual learning outcomes. Following the same specification and using the dummy variable for coming from developing regions as the control variable, we obtain the results in Table A.1 in the Appendix of E-Companions. These

Table 6. Alternative outcome variable test.

Variables	Log No. of Students Surpass Yiben Cutoff		
	(1)	(2)	(3)
Remote $\times$ Developing	0.098** (0.039)	0.078** (0.040)	0.091** (0.040)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	1,967	1,967	1,967
R ²	0.155	0.183	0.191

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

results are highly consistent with the results in Table 5, providing additional support for our findings from the survey and Hypothesis 2.

8 Robustness Checks

In this section, we report the results of several robustness tests. Specifically, we first use the alternative outcome variable, which could be more comprehensive to represent the overall group of students' learning efficiency. Next, we adopt propensity score matching (PSM) to create matched treatment and control groups based on covariates related to cities' educational resources. Then, we conduct a balanced panel analysis in which we only keep schools that report their performance for three consecutive years. In addition, we test an alternative measure of educational resources using a continuous measure, then a robust definition of the treatment group, and finally remote learning days to measure the intensity of the treatment.

8.1 Alternative Outcome Variable

One might suspect that the variable of students scoring above 600 points may not be representative enough to indicate the learning efficiency of all students. To address this suspicion, we employ an alternative outcome variable: the number or percentage of students surpassing the Yiben cutoff score of schools. This threshold signifies the minimum score requisite for "Yiben" university admission. In China, now there are approximately 600 YiBen universities, which means that it is common and easy for students to be admitted to Yiben universities. Thus, we can infer that this outcome variable is more comprehensive to represent the overall group of student's academic performance by counting the number of students admitted by Yiben universities. Table 6 reports the estimated results using the alternative outcome variable. The findings are highly consistent with our main results, which provides additional support. Moreover, we replicate this analysis using the percentage of students surpassing Yiben Cutoff as the dependent variable and the results are highly consistent, presented in Table A.2 in the Appendix of E-Companions.

Table 7. Matching analysis.

Variables	Log No. of Students Above-600		
	(1)	(2)	(3)
<i>Remote × Developing</i>	0.250*** (0.061)	0.322*** (0.057)	0.297*** (0.060)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	1,277	1,277	1,277
R ²	0.252	0.257	0.312

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Furthermore, when focusing on the magnitude of the coefficient of the interaction term, we calculate that the percentage change in the number of students surpassing Yiben cutoff score driven by remote learning is 9.5%.⁸ Recall that in Table 2, the percentage change in the number of students above-600 driven by remote learning is 24.5%. Comparing them, we find that while remote learning keeps showing positive effects in bridging the educational inequality between developed and developing regions, such benefits tend to become weaker when considering the lower-performance students.

8.2 Matching Strategy

To create more comparable units for our DID framework, we conduct the matching on the city level by utilizing covariates related to educational resources. Specifically, we use the local GDP per capita, education expenditure per capita, book collection of public libraries per capita, and the secondary school teacher-to-student ratio of each city in our matching process. Using the data collected from the provincial Statistical Yearbooks, we use PSM to create matched treatment and control groups. We then compared the covariates of both groups by analyzing the mean value of weighted covariates. Table A.3 in the Appendix of E-Companions presents the mean values of each covariate in the two groups and the t -tests conducted to compare them. The t -tests assess whether our matching aligns with the underlying assumption that covariates are balanced between the treatment and control groups after matching. The results in Table A.3 in the Appendix of E-Companions indicate no significant differences in the covariates used for PSM between the treatment and control groups, implying a statistical balance between them. Based on the balanced sample, we re-run the equation (1) and obtain results in Table 7. We find that the results are highly consistent with the prior main results, providing additional support. Additionally, we replicate this analysis using the percentage of students scoring above 600 as the dependent variable and find highly consistent results, shown in Table A.4 in the Appendix of E-Companions.

Table 8. Balanced panel test.

Variables	Log No. of Students Above-600		
	(1)	(2)	(3)
<i>Remote × Developing</i>	0.163*** (0.049)	0.227*** (0.050)	0.191*** (0.048)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	1470	1470	1470
R ²	0.249	0.273	0.305

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

8.3 Balanced Panel Analysis

A potential concern arises from the voluntary nature of disclosing students' performance by some high schools, as there could be a tendency for schools to share their data selectively, particularly when performance is favorable. Consequently, our dataset before and after the implementation of remote learning might consist of dissimilar sets of schools. To mitigate this concern, we employ a balanced panel analysis. This involves constructing a balanced panel by retaining schools that consistently report their performance publicly each year. This meticulous approach ensures a like-for-like comparison, allowing us to assess the learning outcomes of the same cohort of schools both before and after the introduction of remote learning. By adopting this methodology, we can effectively minimize potential biases stemming from variations in school samples during the periods before and after the implementation of remote learning.

Table 8 reports the estimated results using the balanced panel. The findings are consistent with our main results: within the same group of schools, the learning outcomes of students in developing regions increase significantly after remote learning compared to students in developed regions, which means that remote learning effectively reduces educational inequality. We also conduct this analysis using the percentage of students scoring above 600 as the dependent variable as another robust test and find highly consistent results, listed in Table A.5 in the Appendix of E-Companions.

8.4 Measure of Educational Resources

Educational inequality fundamentally arises from the uneven distribution of educational resources. As previously mentioned, we differentiate between developed and developing regions based on the level of available educational resources. In this section, we introduce a set of continuous variables that gauge educational resources to validate the robustness of our findings. The first variable pertains to the average education investment per student within a city, calculated as the local education investment divided by the total number of local high school students. This data is sourced from the provincial Statistical Yearbooks. The second variable focuses on the quantity

Table 9. Measure of educational resources.

Variables	Log No. of Students Above-600			
	(1)	(2)	(3)	(4)
Remote $\times$ Average Education Investment	−0.008*** (0.002)			
Remote $\times$ No. of 985 Universities		−0.121*** (0.024)		
Remote $\times$ No. of 211 Universities			−0.052*** (0.009)	
Remote $\times$ No. of YiBen Universities				−0.019*** (0.005)
Year Fixed Effects	Yes	Yes	Yes	Yes
School Fixed Effects	Yes	Yes	Yes	Yes
Control Variables	Yes	Yes	Yes	Yes
Observations	2,829	2,829	2,829	2,829
R ²	0.272	0.266	0.270	0.264

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). For simplicity, we only report the variables of interest. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

of universities within a city, serving as a strong indicator of the educational resources allocated by the government. In China, universities fall into three main categories: 985 universities, 211 universities, and Yiben universities, each representing distinct tiers of higher education institutions. Consequently, we utilize the counts of these three university types to characterize the educational strength of a given city.

We use these variables as the proxy for the educational resources of cities. We extend equation (1) by replacing the dummy variable that defines the treatment and control groups with the continuous variables that measure the level of educational resources above. In this way, we relax the classic DID specification to the generalized DID specification by Angrist and Pischke (2008: 229). Table 9 reports the estimation results. We find the same results as before: the gap in students' learning outcomes across cities with more and less educational resources has significantly reduced after adopting remote learning. We also conduct this analysis using the percentage of students scoring above 600 as the dependent variable as another robust test and find highly consistent results, displayed in Table A.6 in the Appendix of E-Companions.

8.5 Measure of Developing Regions

Recall that we define the provincial capital cities as developed regions. This definition might misclassify some cities with

Table 10. Developing regions measure.

Variables	Log No. of Students Above-600		
	(1)	(2)	(3)
Remote $\times$ Developing	0.175*** (0.047)	0.230*** (0.048)	0.188*** (0.046)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	2,829	2,829	2,829
R ²	0.250	0.242	0.284

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

advanced development. We address this concern by reclassifying cities in developing regions according to the economic development plan.⁹ For example, Shenzhen and Qingdao have similar or even higher economic (GDP) and population growth as provincial capital cities. We reclassify these cities also as developed regions and repeat our DID analysis using equation (1). We find highly consistent results shown in Table 10. We also conduct this analysis using the percentage of students scoring above 600 as the dependent variable as another robust test and find highly consistent results, shown in Table A.7 in the Appendix of E-Companions.

8.6 Remote Learning Days Measure

Recall that we use a dummy variable to indicate before and after remote learning. In this section, we replace this dummy variable with a continuous variable, the remote learning duration, which measures remote learning intensity to confirm the robustness of our results. Starting in January 2020, students were required to adopt remote learning at home. After COVID-19 was effectively controlled, the local provincial governments released policies to let students return to campus to take in-person classes. All high schools in a province reopened on the same date. Thus, we collect the exact reopening date of high schools in each province. We then construct the variable *Remote Days*, which measures the total number of days that local students participate in remote learning. As mentioned earlier, the NCEE in 2020 was postponed by one month and thus, the increase in remote learning days in 2020 is not equal to the decrease in in-person teaching days in our study.

We extend equation (1) by replacing the *Remote* dummy with *Remote Days*. We relax the classic DID specification to the generalized DID specification in Angrist and Pischke (2008: 229), which allows us to observe the effect of remote learning length on educational inequality. We expect that when remote learning days are longer, the effect of remote learning on reducing educational inequality will be stronger. Table 11 reports the estimation results. We find results consistent as before: the coefficients of *Remote Days* $\times$ *Developing* are significant and positive, meaning that the intensity of remote learning plays a big role in improving learning outcomes.

Table 11. Remote learning days measure.

Variables	Log No. of Students Above-600		
	(1)	(2)	(3)
<i>Remote Days</i> × <i>Developing</i>	0.004*** (0.001)	0.005*** (0.001)	0.004*** (0.001)
<i>Remote Days</i>	0.003*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
Year Fixed Effects	No	Yes	Yes
School Fixed Effects	Yes	No	Yes
Control Variables	Yes	Yes	Yes
Observations	2,829	2,829	2,829
R ²	0.254	0.245	0.288

Note: This table reports the estimated coefficients and robust standard errors (in parentheses). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

We also conduct this analysis using the percentage of students scoring above 600 as the dependent variable as another robust test and find highly consistent results, demonstrated in Table A.8 in the Appendix of E-Companions.

9 Concluding Remarks

This study investigates the impact of remote learning on reducing educational inequality between developed and developing regions. We utilize data from students' performances on the college entrance exam for three years and find that remote learning increases the average number of students scoring above 600 points in schools located in developing regions by 24.5% compared to those in developed regions. We conducted a robustness analysis using various alternative measures, which further confirmed the validity of our findings. Moreover, we discovered that the effect of remote learning is contingent on the digital capabilities of schools. Schools with stronger digital capabilities can better improve students' learning outcomes through remote learning and thus more effectively reduce educational inequality. Finally, we conducted a survey to explore the mechanisms behind our findings, which provides additional support for our main and moderating results.

The study's findings bear significant implications for the research community. While previous investigations have predominantly focused on the impact of remote learning on academic performance, scant attention has been directed towards its potential to alleviate educational inequality. Remote learning possesses distinctive attributes that hold promise in mitigating disparities in education. For instance, it enables students to transcend the confines of their own institutions, facilitating interaction with peers and instructors from diverse regions while also allowing personalized learning. These aspects collectively foster a more equitable learning environment, particularly benefiting students confronted with disadvantages, thereby contributing to the attenuation of educational inequality across regions. Consequently, remote learning has garnered increasing attention in recent years due to its capacity to

address prevailing societal issues within educational systems. However, we also find that the benefits of remote learning in reducing educational inequality become weaker when considering the lower-performance students. Furthermore, in the era of digitization, the efficacy of remote learning might be constrained if the existing information technology infrastructure disparities across regions are overlooked. This study highlights the crucial role of the digital divide in this context. By incorporating this dimension, our research expands the horizons of remote learning investigation, offering a fresh vantage point on its potential to address evolving societal challenges within the realm of education.

Our study holds significant implications for policymakers, shedding light on the transformative potential of remote learning in addressing educational inequality. The empirical evidence we present underscores that remote learning can serve as a potent tool for diminishing disparities by granting students in developing regions access to valuable educational content, instructors, and peer interactions. Importantly, our findings underscore the pivotal role of digital capabilities in unlocking the full benefits of remote learning. We believe our findings can enhance policymakers' understanding of the value remote learning brings to mitigating educational inequality. In response to our findings, policymakers could consider integrating remote learning as a supplementary tool within traditional classroom settings. By doing so, they can facilitate broader access to quality educational resources, benefiting students regardless of their geographic location. Furthermore, advocating for the expansion of online educational platforms and bolstering technological infrastructure can empower individuals with the digital skills necessary for a seamless remote learning experience. Ultimately, addressing the digital divide across regions can amplify the positive impact of remote learning in rectifying existing disparities in educational resources. This, in turn, could significantly contribute to the reduction of educational inequality and the promotion of social and economic mobility on a larger scale.

Our study presents several limitations that warrant consideration. First, our investigation exclusively centered on students within the Chinese context, potentially limiting the generalizability of our findings to other geographical settings. To gain a more comprehensive understanding, future research endeavors could extend their focus to diverse contexts to discern if similar outcomes prevail. Second, our analysis hinged upon school-level data to evaluate students' learning outcomes. To provide a more granular perspective, upcoming studies could amass performance data at the individual level. This approach would facilitate the examination of performance distribution alongside pertinent student-level attributes, thereby affording a more nuanced comprehension of the underlying dynamics. Third, due to the data constraints, we use the number of students above-600 to represent high schools' achievement in NCEE. Future studies could consider using alternative outcome variables, such as the school's

college admission results, to obtain a more comprehensive understanding. Lastly, since our sample includes schools having students above-600, some may argue that it is less likely to represent low-ranking high schools in each region (i.e., city/province). Although we acknowledge this potential limitation, it may not undermine the robustness of our findings for two reasons. First, our focus is on educational inequality between developed and developing regions. Therefore, even though our sample is more likely to represent higher-ranking high schools in each region, the findings in this paper still reflect this type of educational inequality to some extent. Second, when high- and low-ranking high schools are located within the same city, they are likely to be similarly affected by certain factors, as they could be governed by the same local policies and resources. It is reasonable to assume that the impact of remote learning, through its connectivity and flexibility, on high-ranking schools could be similar to its impact on low-ranking schools in the same region. Thus, employing low-ranking high schools as the sample might yield findings akin to those presented in this paper. However, future studies that can access data on the educational outcomes of low-ranking high schools across different regions could apply our research framework to find out whether the results remain consistent.

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Notes

1. Source: <http://2015top.yuantiku.com/#/index>, accessed March 2024.
2. Note that while these results demonstrate robustness among the top 10% of students in the country, it's pertinent to recognize that these findings may not translate similarly for, for example, the bottom 50% of students in the NCEE.
3. Since we use the capital city in a province as the control group (i.e., developed regions) and the non-capital cities in a province as the treatment group (i.e., developing regions), we exclude the cities of *Beijing*, *Shanghai*, *Tianjin*, and *Chongqing*, which are major cities that do not belong to any province. Moreover, the national college entrance examination systems in the *Jiangsu* and *Zhejiang* provinces are significantly different from those in other provinces

in China in that they design the entrance exam independently, and their total scores are different from other provinces in our sample. We, therefore, exclude these two provinces. Furthermore, we do not find much information about high schools' announcements in *Qinghai*, *Xinjiang*, *Ningxia*, *Hainan*, and *Xizang*, and thus we also exclude these provinces.

4. Source: <http://2015top.yuantiku.com/#/national>, accessed March 2024.
5. Source: http://www.xinhuanet.com/politics/2020-03/31/c_1125793628.htm, accessed March 2024.
6. Source: http://www.moe.gov.cn/jyb_xwfb/moe_2082/zl_2019n/2019_zl96/202001/t20200103_414320.html, accessed March 2024.
7. Because the outcome variable is logged, the percentage change in the outcome variable is computed as $e^{0.219} - 1 = 24.5\%$.
8. Again, since the outcome variable is logged, the percentage change in the outcome variable is $e^{0.091} - 1 = 9.5\%$.
9. Cities included in this plan possess economic management authority similar to that of provincial capital cities, which is crucial for attracting people to settle and the development of their local economy.

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